

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Designed and implemented **scalable data integration solutions** using **AWS EMR, S3, AWS Glue, and Lambda** to extract, transform, and distribute payment transaction data across disparate banking systems, reducing data reconciliation time by **40%** and enabling real-time operational visibility for cross-functional teams.
- Built and optimized **PySpark data pipelines** processing **500M+ daily payment messages** (SWIFT MT/MX formats) using **Apache Airflow** for orchestration, implementing Infrastructure as Code via **CloudFormation** and achieving **99.95% pipeline reliability** in production environments.
- Delivered **CI/CD automation** using **Jenkins, Git, Maven, and Artifactory** for cloud deployments across AWS services, enabling **15+ daily releases** with zero-downtime deployments and reducing deployment failure rate to **2%**.
- Managed data warehousing and modeling for financial services analytics using **AWS RDS (PostgreSQL/Oracle), Databricks, and Athena** with **Parquet, ORC, Avro, and Delta Lake table formats**, supporting P1/P2 SLA-driven incident response and regulatory compliance (GDPR, ISO 20022 workflows) across interconnected enterprise systems.
- Engineered **complex data acquisition and integration logic** to consolidate disparate payment operations systems (ERP modules, SEPA/cross-border platforms, tokenization workflows) into a unified data lake, reducing manual reconciliation overhead by **35%** and enabling business intelligence teams to deliver stakeholder-facing dashboards within **48 hours**.
- Collaborated with Product Management, Integration Engineering, Quality Engineering, and remote geographically distributed System Admin teams to define data requirements, troubleshoot distributed systems issues, and communicate complex data architecture concepts to non-technical stakeholders, ensuring **100% UAT sign-off** and smooth post-implementation triage.

SKILLS

Big Data & Data Engineering – Apache Spark, PySpark, Scala, Apache Airflow, AWS EMR, AWS Glue, ETL/ELT pipeline design and optimization, data acquisition frameworks.

AWS Data Warehousing & Cloud Platforms – AWS S3, AWS Athena, AWS Lambda, AWS RDS, EMR, Glue, CloudFormation, Terraform, Infrastructure as Code (IaC), AWS CloudWatch.

Data Integration & Storage Formats – Parquet, ORC, Avro, Delta Lake, Iceberg table formats, data modeling, data warehousing, data lake architecture, data security concepts.

CI/CD & Build Automation – Jenkins pipelines, Git version control, Maven, Artifactory, YAML configuration, cloud deployment automation, change-control management.

Data Querying & Analytics Platforms – Databricks, Amazon Athena, RDS (PostgreSQL, MySQL, Oracle), NoSQL (DynamoDB), ad-hoc SQL queries, statistical analysis, data validation.

Programming & Backend Development – Python (pandas, NumPy, scikit-learn), Java, Spring Boot, REST API design and integration, distributed systems troubleshooting, problem-solving.

Financial Services & Domain Knowledge – SWIFT messaging (MT/MX formats), ISO 20022 message flows, SEPA/cross-border payments, PSD2/Open Banking, payments processing, GDPR and regulatory compliance, enterprise ERP integrations.

EDUCATION AND TRAINING

**MASTER OF SCIENCE in Data Analytics – National College of
Ireland, Dublin**

02/2026

**BACHELOR OF ENGINEERING in Computer Science – SNS College
of Technology, Coimbatore, India**

01/2021

PROJECTS

Workforce Data Lake & Analytics Integration Platform (HCL Technologies)

- Architected and delivered a **petabyte-scale enterprise data lake** on **AWS S3, Glue, and EMR** consolidating workforce data from 8+ disparate HR, payroll, and financial systems using **Apache Spark (PySpark)** and **Delta Lake** table formats, enabling HR analytics and business intelligence teams to build self-service dashboards and reduce query latency by **60%**.
- Implemented **Apache Airflow DAGs** orchestrating **30+ scheduled data pipelines** that ingested, transformed, and distributed **15M+ employee records daily** across **Parquet, ORC, and Avro formats**, achieving **99.9% on-time completion** and automatic failure remediation using retry logic and CI/CD triggered redeployment.
- Built Infrastructure as Code templates using **CloudFormation and Terraform** to provision **auto-scaling EMR clusters, Lambda functions for micro-ETL jobs, and RDS PostgreSQL data warehouses**, reducing infrastructure provisioning time from **3 weeks to 2 days** and enabling development teams to spin up isolated analytics environments on-demand.
- Designed and deployed **data security, encryption-at-rest (S3 KMS), and row-level access control** mechanisms to ensure GDPR compliance and regulatory audit trails across all data lakes and warehouses, maintaining **zero security incidents** and passing **100% of compliance audits**.
- Led cross-functional collaboration with Product Management, Quality Engineering, and remote System Admin teams to define data SLAs, troubleshoot distributed system bottlenecks, and document complex data flows; delivered **50+ knowledge-transfer sessions** enabling non-technical stakeholders to consume data assets via Athena and Databricks notebooks.

SWIFT Payments Operations & Reconciliation Data Pipeline (HCL Technologies)

- Engineered a real-time **data integration pipeline** using **Python, AWS Lambda, and S3** to consume **SWIFT MT/MX message streams** from 10+ banking channels, normalize **ISO 20022 message payloads**, and distribute transaction records across **Oracle RDS and Databricks** for payments reconciliation and regulatory reporting, supporting **5M+ daily cross-border payment messages** with **99.95% delivery SLA**.
- Built **Apache Airflow workflows** orchestrating daily reconciliation jobs that compared transaction states across SEPA/instant payment systems and legacy ERP platforms, identifying discrepancies within **15 minutes** and auto-generating alerts for P1/P2 incident triage, reducing manual reconciliation effort by **45%** and regulatory remediation time by **30%**.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)